Course Code: CMP 234 | Full Marks: 100
Course title: Computer Graphics (3-1-2) | Pass Marks: 45
Nature of the Course: Theory and Practical | Total Lectures: 45 hours

Specific Objectives	Contents
Provides a brief introduction of the field of Computer Graphics and familiarize students with basic graphics hardware and software. It intends to enhance the understanding of the technology used for producing graphical output on display devices.	**Unit 1: Overview of Computer Graphics and Graphics Systems (7 hrs)** 1.1 Introduction, Applications and Recent trends of Computer Graphics 1.2 Interactive Input Devices 1.3 Display Devices, Color Monitors, Hard Copy Devices 1.4 Raster and Random Scan Systems and Architectures 1.5 Video Controller 1.6 Use of Digital to Analog Converter and Frame Buffer Organization 1.7 Graphics Software, Modern Graphics Hardware (GPU)
Learn about various scan conversion and area filling algorithms. Students will be able to implement these scan conversion algorithms using a programming language. This chapter helps students learn how algorithms are used to generate output primitives and fill them with specified intensities.	**Unit 2: Scan Conversion Algorithms (7 hrs)** 2.1 Scan Conversion 2.2 Line Drawing Algorithms 2.2.1 Digital Differential Analyzer 2.3.2 Bresenham's Algorithm 2.3 Circle Generation Algorithm 2.4 Ellipse Generation Algorithm 2.5 Filled Area Primitives 2.5.1 Scan Line Polygon Fill Algorithm **2.5.2** Boundary Fill Algorithm, Flood Fill Algorithm
Get students well acquainted with the two-dimensional geometric transformations and viewing Able to make use of various transformation operations and viewing operations to reposition and resize objects in two dimensional scene Learn about the operations used in viewing routines that convert a world coordinate scene description to a display for an output device.	**Unit 3: Two Dimensional Geometric Transformations and Viewing (8 hrs)** 3.1 Two Dimensional Transformations 3.1.1 Translation, Rotation, Scaling, Shearing, Reflection 3.2 Matrix Representations of Transformations 3.3 Homogeneous Coordinate System 3.4 Composite Transformations 3.5 Windowing Concepts, Two Dimensional Viewing Pipeline 3.6 Window to Viewport Transformation 3.7 Line Clipping Algorithm: Cohen-Sutherland **3.8** Polygon Clipping: Sutherland-Hodgeman

Learn about the extensions of two-dimensional methods for geometric transformation in three dimensions. Helps them apply the knowledge of three dimensional rendering and transformations for representing objects more accurately	**Unit 4: Graphics in Three Dimensions (8 hrs)** 4.1 Three Dimensional Coordinate Systems 4.2 Three Dimensional Transformations 4.2.1 Translation, Rotation, Scaling, Reflection 4.3 Three Dimensional Representations 4.3.1 Polygon Surfaces 4.3.2 Cubic Spline and Beizer Curve 4.3.3 Non-Planer Surface: Bezier Surface 4.3.4 Fractal Geometry Method 4.4 Three Dimensional Viewing Transformation and Pipeline 4.5 Projections 4.5.1 Parallel Projection 4.5.2 Perspective Projection **4.6** Clipping in Three Dimensions
Get acquainted with the techniques used for bringing visual realism in computer generated graphical output primitives. Familiarize with visible surface detection techniques for creating realistic displays and in computing	**Unit 5: Visual Realism (7 hrs)** 5.1 Hidden Surfaces and Hidden Surface Removal Approaches 5.1.1 Back-Face Detection 5.1.2 Depth Buffer Method 5.1.3 A- buffer method 5.1.4 Scan Line Method **5.1.5** Depth Sorting Method
light intensities of surfaces using lighting or shading models for generating realistic views	**5.2 Illumination and Shading Methods** 5.2.1 Illumination Theory and Models 5.2.1.1 Ambient Light 5.2.1.2 Diffuse Reflection 5.2.1.3 Specular Reflection (Phong Model) 5.2.2 Polygon Surface Shading Methods 5.2.2.1 Constant Shading (Flat Shading) 5.2.2.2 Gouraud Shading 5.2.2.3 Phong Shading 5.2.2.4 Fast Phong Shading **5.3 Color Models** 5.3.1 RGB **5.3.2** CMYK
Gain a good understanding of the use of graphical standards necessary for achieving software portability. Become familiar with the basic structure or format of a graphical file and know the ways of visually representing data sets. Learn about writing graphical programs that can be ported to multiple hardware platforms without extensive rewriting of the programs through the use of graphical standards.	**Unit 6: Graphical Standards (4 Hrs)** 6.1 Need for Machine Independent Graphical Languages 6.2 Graphical Standards: PHIGS, GKS 6.3 Graphics Software Standards and Language Binding 6.4 Overview of Graphics File Formats **6.5** Visualization of Data Sets

Learn about the library functions provided in OpenGL for graphical rendering and performing geometric transformations. Enable students to write programs using functions in OpenGL library for drawing graphical output primitives and performing basic transformations	**Unit 7: Introduction to OpenGL (4 Hrs)** 7.1 Overview and Basic Architecture of OpenGL 7.2 GL Related Libraries 7.2.1 Drawing Basic Output Primitives and Polygons 7.2.2 Call Back Functions and Input Handling 7.2.3 Basic Transformations **7.2.4** Projections and Lighting

Pokhara University Faculty of Science and Technology	
Course Code: CMM 320(3 Credit)	Full Marks: 100
Course Title: Signal, System and Processing (3-0-2)	Pass Mark: 45
Nature of the Course: Theory and Practice	Total Lectures: 45 hours
Level: Bachelor/ Year: III/ Semester: V	Program: Bachelor of Engineering in Information Technology

1. Course Description:

"Signal System and Processing" is a fundamental aspect of Information Technology Engineering. This course starts from the basic concepts of continuous and discrete-time signals and proceed to learn how to analyze data via the Fourier transform, how to manipulate data via digital filters and how to convert analog signals into digital. The solid theoretical bases are complemented by applied examples in Matlab. Design and lab exercises are also significant components of the course.

2. General Objectives:

a) To provide fundamental knowledge of digital signal processing techniques and applications.

b) To describe signals mathematically and understand how to perform mathematical operations on signals.

c) To learn basic methods of spectral analysis

d) It will provide knowledge and realization of Digital filter.

e) Understand the principle of Linear System and digital signal processing fundamentals.

3. Methods of Instructions:

- lecture, tutorial ,discussion, and practicals.

4. Contents in Detail	
Specific Objectives	**Contents**
• Learn about the basics of discrete time signals and its classification • Understand the concept of different types of discrete time systems • Comprehend the concept of discrete linear convolution and its properties	**1. Introduction to discrete time signals and systems: (7hrs)** ● Signal(continuous time and discrete time) and its types, , Basic Elementary signals - unit impulse signal, unit step ,signal, ramp signal , Sigmoid signal, exponential signal, sinusoidal signal, Basic Elements of DSP system, application of DSP ● Basic signal transformation techniques- Time shifting, Amplitude scaling, Time scaling, Time inversion and Combined Operations ● Analysis of signals using discrete time Fourier series and Discrete time Fourier Transform. ● Discrete time system, Classification- static and dynamic systems, time variant and time invariant systems, linear and nonlinear systems, causal and non-causal systems, stable and unstable systems ● Linear Time Invariant (LTI) System, Convolution summation of discrete systems, response to discrete inputs, Properties of Convolution – Commutative property, Associative property, Distributive property, Causality and Stability of LTI system
• Comprehend the concept of Z-Transform, ROC and properties of Z-transform • Learn to perform forward and inverse Z-transform • Understand the difference between Unilateral and Bilateral Z-transform • Grasp the concept of causality and stability	**2. Z-transform (7Hrs)** ● Z-transform and Region of Convergence ● Left-sided, right sided and two-sided sequences and its z-transform, relationship to causality. ● Z-transform properties- linearity, time shifting, scaling , differentiation, convolution, Parseval's theorem ● Inverse Z-transform using long division and partial fraction expansion
• Grasp the concept of plotting Magnitude and Phase Response of LTI	**3. Analysis of LTI system in frequency domain (4hrs)** ● Frequency response of LTI system, response to complex exponential.

system • Understand the concept of poles and zeros and their relationship to causality	● Linear constant coefficient, difference equation and corresponding system function ● Relationship of frequency response to pole zero of system.
• Learn to represent FIR and IIR filters in direct, cascade, parallel and lattice structure. • Realization of hardware implementation of the filter by studying the filter elements like adder, delay element, multiplier etc.	**4. Discrete filter structures (7 hrs)** ● FIR filter ● Structures of FIR filter o Direct form o Cascade form o Lattice realization ● IIR filter ● Structures of IIR filter o Direct form I and II o Cascade form o Parallel form o Lattice and Lattice ladder structures
• Grasp the concept of designing analog Butterworth and Chebyshev filters. • Comprehend how analog IIR filter is converted to digital IIR filter • Understand the idea of converting LPF prototype filter to LP, HP, BP and BS filters • Learn the property of Butterworth, Chebyshev and Elliptical filters and their differences.	**5. IIR Filter Design (6 hrs)** ● IIR filter design by transformation o Impulse-invariance method o Bilinear transformation ● Properties and types of digital low pass filter o Butterworth filter o Chebyshev filter o Elliptical filter ● Basic Analog Filter approximation and its digital design o Butterworth Approximation
• Understand Gibbs phenomena • Grasp the concept of designing FIR filters using Windowing and frequency sampling methods. • Understand the concept of designing optimum FIR	**6. FIR Filter Design (6 hrs)** ● FIR filter design using windows o Rectangular window and its magnitude response, Gibbs phenomenon o Hanning window o Hamming window ● Comparision of different types of window

filters	● Linear phase FIR filter(symmetric and ant symmetric FIR filter) ● FIR filters design using Frequency sampling method.

• Understand the concept of Discrete Fourier Transform and its properties
• Grasp the concept of circular convolution
• Learn about FFT, DIT, DIF algorithms.
• Comprehend the computational complexity of FFT algorithm
	7. The Discrete Fourier Transform (8 hrs)
	● The discrete Fourier transforms(DFT) and its computation, Inverse DFT
	  o Classical approach and matrix method
	● Properties of the DFT
	  o Linearity
	  o Time Shifting
	  o Frequency Shifting
	  o Conjugation
	  o Duality
	  o Convolution
	● Circular convolution
	● Fast Fourier Transform(FFT) and its computation using radix-2 algorithm
	  o Decimation in Time(DIT)
	  o Decimation in Frequency(DIF)
	● Relationship with DTFT

5. List of Tutorials The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover all the required contents of this course:	
SN	**Tutorials**
1.	Study of basic analog and digital signals and their transformation
2.	Plot of a Continuous and discrete time signal , numerical on their analysis, system properties
3.	Analysis of a LTI system using pole zero diagram and determine the stability of that system.
4.	Solve the problems related to Discrete Filter
5.	Realization of FIR and IIR filter using filter structures
6.	Design of IIR filter and FIR filter using various methods.
7.	Analysis of a digital signal using Discrete Fourier Transform

6. List of Practicals(30 hours for a group of maximum 24 students)	
SN	**Implementation Description**
1	Generate and Investigate Basic Continuous and Discrete Time Signals i. Unit impulse signal, Unit step signal, Ramp signal, Sinusoidal Signal, Exponential signal and square signal ii. Compute even and odd parts of signal iii. Convolution Sum of two sequences
2.	Frequency response and pole zero plot of differential equation
3.	Compute 4-point and 8-point DFT using FFT and investigate their frequency responses
4.	Design an IIR lowpass filter using Impulse Invariance and Bilinear Transformation Method
5.	Design a FIR filter using different windows and compare the result.
6.	Real-World Digital Signal Analysis and Processing examples and demonstrations

7. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**30**	**Semester End**	**50**
Attendance &Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		**20**		
Attendance &Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		**50**		
Full Marks: 50 + 50 = 100				

Students' Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References
Text Books: Alan V. Oppenheim, Ronald W. Schafer "Discrete-Time Signal Processing", Prentice Hall
References: **1.** J.G. Proakis and D.G. Manolakis, Digital Signal Processing Prentice Hall of India. **2.** S.K. Mitra, Digital Signal Processing. A Computer based approach, Mc Graw Hill.

Pokhara University

Faculty of Science and Technology

Course No.: CMP 333	Full marks: 100
Course title: Multimedia Systems (2-1-2)	Pass marks: 45
Nature of the course: Theory and Practical	Total Lectures: 30 hrs
Level: Bachelor	Program: BE (IT)

Authors

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1. Course Description

This course introduces students to the key concepts and technologies behind multimedia, including sound, images, video, animations and data compression. Students will learn both the theory and practical skills needed to design and work with multimedia systems. Through a mix of lectures and hands-on labs, students will gain the ability to create and evaluate multimedia content, preparing you to apply these skills in real-world settings.

2. General Objectives

- Foundational understanding of multimedia systems.
- Development of technical proficiency in multimedia processing.
- Application of knowledge in designing multimedia components.
- Encouragement of critical thinking in multimedia evaluation.

3. Methods of Instruction

Lectures for theoretical foundation.

Discussion Sessions for interactive learning.

Hands-on **Practical Work** for applied skills.

Project-Based Learning for integrative experience.

Readings and **Assignments** for reinforcement and assessment.

4. Contents in Detail

Specific Objectives	Contents
Identify and describe the essential components of multimedia systems, including text, audio, images, and video, and understand their roles and interactions.	**1. Introduction to Multimedia(3 hrs)** 1.1. Overview of Multimedia, Building Blocks of multimedia 1.1.1. The Medium aspects, Application areas and usefulness,Properties of multimedia 1.1.2. Global Structure of Multimedia 1.1.3. Data Stream Characteristics: Asynchronous and Synchronous Transmission Mode
Describe how sound and music are represented in digital systems, including concepts of sampling, bit depth, and MIDI technology. Outline the basic concepts of image processing, including image synthesis, analysis, and enhancement techniques.	**2. Sound and Graphics (5 hrs)** 2.1. Basic Sound Concepts 2.1.1. Computer Representation of Sound/Audio 2.1.2. Computer Representation of Music: MIDI 2.1.3. MIDI Devices, MIDI Modes, MIDI Messages and MIDI Software. 2.2. Speech Concepts 2.2.1. Speech Processing and Speech Analysis 2.2.2. Speech Recognition 2.3. Image Definition: Basic Concepts 2.3.1. Image Processing Fundamentals(Image Synthesis,Image Analysis, Image Transmission) 2.3.2. Image Recognition Steps

	2.3.3. Image Enhancement Methods
Understand how video is represented and formatted, and explain the significance of different video formats and Compare and contrast computer-based animation, handwritten animation, and stop-motion animation, including their techniques and applications.	**3. Video and Animation Technology (3 hrs)** 3.1. Basic Video Concept: Representation and Format 3.2 Television: High Definition Television(HDTV) 3.3 Animation: Types of Animation 3.3.1 Computer Based Animation 3.3.2 Handwritten Animation 3.3.3 Stop Motion Animation 3.4 Design Principles of Animation, Methods of Controlling Animations

Explain the differences between lossy and lossless data compression methods, including examples and applications of each and describe the principles and applications of key data compression techniques such as Huffman coding and JPEG compression.	**4. Data Compression Methods(6 hrs)** 4.1. Coding Requirement 4.2. Types of Compression: Lossy and Lossless Compression 4.2.1. Run length Coding,Entropy Coding 4.2.2. Huffman Coding 4.2.3. Arithmetic Coding 4.3. Coding Standards 4.3.1. JPEG: JPEG compression steps 4.3.2. MPEG: Evolution of MPEG, Data Stream of MPEG,MPEG Audio 4.4. Digital Video Interactive (DVT), h.261(p✕64)
• Understand the fundamental principles of optical storage media, including CD DA and CD-ROM technologies, and their features.and Outline the various DVD formats and their uses, including differences from other optical storage media.	**5. Storage Media (3 hrs)** 5.1. Basic of Optical Storage Technology 5.1.1. Compact Disk Digital Audio (CD DA):Audio Data Rate,Capacity, Error handling, Eight to Fourteen Modulation 5.1.2. Compact Disk Read Only Memory(CD-ROM):Blocks,CD-ROM modes 5.1.3. CD-ROM Extended Architecture (CD-ROM/XA): Form 1 and Form 2 5.2. DVD, DVD Formats

Describe the principles of document architecture, focusing on SGML and ODA standards. and define hypertext and hypermedia, and understand the basic architecture and components of hypermedia systems, including nodes and pointers.	**6. Document Architecture,Hypertext and MHEG (4hrs)** 6.1. Documents, Document Architecture :SGML and ODA 6.2. Hypertext, Hypermedia and Multimedia 6.3. Hypermedia System: Architecture, Nodes and Pointers 6.4. Brief History of Hypertext and application areas of Hypertext. 6.5. MHEG: Class Hierarchy of MHEG objects, MHEG Identifier and MHEG Descriptor, Contents, Behaviour, Interaction,Container and Closing Comments.

Explain the concepts of real-time operating systems and resource management for multimedia applications, including scheduling methods like EDF and RMA. Outline the components and functions of multimedia communication systems, including session management and quality of service (QoS) principles	**7. Multimedia Operating System and Communication Systems (6 hrs)** 7.1. Introduction to MOS, Real Time Systems and Deadline 7.1.1. Resource Management: Resource and Requirements 7.1.2. Phases of Resource Reservation and Management Process, Allocation Scheme 7.1.3. Continuous Media Resource Model 7.1.4. Process Management, Real Time Processing Requirement, Traditional Real Time Scheduling(EDF and RMA) 7.2 Communication Architecture:Application Subsystem and Transport Subsystem 7.2.1 Session Management, Multimedia Workstation 7.2.2 Quality of Service and Resource Management, Design Principle of QoS.

5. Practical Works

Laboratory work of 30 hours per group of maximum 24 students should cover implementation of the following

SN	Implementation Description
1	Implementation of basic sound processing techniques, including audio synthesis and manipulation.
2	Development of a simple MIDI-based music creation tool that supports basic MIDI messages and device integration
3	Image processing fundamentals: Implementing techniques for image enhancement, including point processing and spatial filtering.
4	Video encoding and compression using basic standards such as JPEG and MPEG, focusing on practical application.
5	Creation of a basic animation sequence using keyframe animation, demonstrating principles of motion and timing.
6	Implementation of a simple multimedia application that integrates sound, image, and video elements.
7	Data compression techniques: Practical implementation of lossless compression methods such as Huffman coding.
8	Storage media exploration: Demonstration of DVD, HDD and SSD data storage and retrieval methods.
9	Design and development of a simple multimedia document using SGML or a similar markup language.
10	A final project that integrates various multimedia elements learned in the course, such as combining sound, image, and video into a cohesive system.

For the practical works, we may use several key Python libraries to manage different multimedia tasks. **NumPy** and **SciPy** will help us with the basics of scientific computing and image processing. **Pillow** and **OpenCV** will be our go-to for working with images and videos. To handle audio, use **PyDub** and **Librosa**, while **Matplotlib** will be handy for creating visualizations. **MoviePy** will make video editing straightforward, and **PyGame** will be great for building multimedia apps. We may use **SoundFile** for audio file handling, and **FFmpeg**

(with ffmpy) for advanced audio and video processing tasks. These tools will be essential as you dive into the practical side of multimedia systems.

Students should submit a project work that uses all the knowledge obtained from this course to solve any problem chosen by themselves. The marks for the practical evaluation must be based on the project work submitted by students.

6. List of Tutorials

The various tutorial activities that suit your course should cover all the content of the course to give students a space to engage more actively with the course content in the presence of the instructor. Students should submit tutorials as assignments or class works to the instructor for evaluation. The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover the content of this course:

A. Discussion-based Tutorials: (3 hrs)

i. Discussion on the importance and applications of multimedia in modern technology.
(Class discussion)

ii. Analysis of different multimedia systems and their real-world use cases.
(Class discussion)

iii. Exploring the future trends (like metaverse) in multimedia technology.
(Class discussion)

B. Problem solving-based Tutorials: (6 hrs)

i. Design a system to encode and decode simple audio and video data.

ii. Design a system that implements basic image processing, such as filtering and enhancement.

iii. Suppose you are given a task to design a multimedia system for an educational platform—outline your approach.

C. Review and Question/Answer-based Tutorials: (6 hrs)

i. Case study on the history of multimedia technology and its evolution
(Oral Presentation in class).

ii. Case study on a multimedia system currently used in a specific industry (e.g., entertainment, education, or healthcare).

iii. Students ask questions within the course content, discuss assignments, and review key concepts in preparation for tests or exams.

7. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, internal assessment, lab reports, project works etc. The internal evaluation scheme for this course is as follows:

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		30	Semester-End examination	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

Student Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Books

1. Steinmetz, R., Nahrstedt, K., "Multimedia Systems", Springer.

References Books

1. Halsall, F., "Multimedia Communications: Applications, Networks, Protocols and Standards", Pearson.

2. Parekh, R., "Principles of Multimedia", Tata McGraw-Hill.

3. Andleigh, P., Thakrar, K., "Multimedia Systems Design", Prentice Hall.

Pokhara University
Faculty of Science and Technology

Course Code.: MGT 332 | Full marks: 100
Course title: Entrepreneurship and Professional Practice (2-1-0) | Pass marks: 45
Nature of the course: Theory and Practical | Time per period: 1 hour
Total periods: 30
Level: Bachelor | Program: BE

1. Course Description

This course provides an understanding of entrepreneurship and professional practices with the intention to prepare students for entrepreneurial and professional roles. The course covers the fundamentals of entrepreneurship, business process issues, business models and professional ethics and legal issues. The course delivery consists of lecture sessions, guest entrepreneur sessions, case studies, business model and plan development and students' experiential exercises. At the end, students will be able to apply course knowledge on developing a technology-based business model and be prepared for professional roles.

2. General Objective

The course is designed to familiarize entrepreneurial and professional career opportunities in ICT including with more focus to enhance their entrepreneurial attitude and intention. It helps to make students able to understand the foundational issues of entrepreneurship, creativity, innovation and ideation, the process of startup and make students able to develop and present business canvas, plan and deck. It also helps students to understand issues on professional practices and legal issues for the ICT based engineering career.

3. Contents in Detail

Specific Objectives	Contents
To enable students to explain foundational entrepreneurship concepts and familiarize them with entrepreneurial and professional career opportunities	**Unit I: Introduction to Entrepreneurship (4 Hrs.)** 1.1 Fundamentals of Entrepreneurship 1.1.1 Definition of entrepreneurship 1.1.2 Key terminologies: startup, enterprise, entrepreneur, entrepreneurial attitude, intention, and behavior 1.1.3 Comparison: entrepreneurship vs employment 1.1.4 Types of entrepreneurs and their characteristics 1.1.5 Enterprise lifecycle 1.1.6 Role, importance and challenges 1.2 Emerging technologies, trends of global and local startups
To make the students familiar with creativity, innovation and idea generation for	**Unit II: Creativity, Innovation and MVP (3 Hrs.)** 2.1 Concept of creativity, innovation and opportunity recognition 2.2 Sources of new business ideas and tools for idea generation 2.3 Types of innovation: product, process, and business model 2.4 Value proposition and minimum viable product (MVP)

developing viable product	
To make the students analyze startup foundation issues related to environment, competition, market, strategy, team and finance	**Unit III: Startup Foundational Analysis (8 Hrs.)** 3.1 Environmental analysis 3.1.1 Entrepreneurship ecosystem 3.1.2 Entrepreneurship supports and barriers 3.2 Competitor analysis 3.2.1 Identifying competitors 3.2.2 Sources of competitive advantage 3.3 Market analysis and plan 3.3.1 Market research, target customer profile 3.3.2 Key terminologies: SAM (Serviceable Available Market), TAM (Total Addressable Market), SOM (Serviceable Obtainable Market), and GTM (Go-To-Market) strategy 3.3.3 Branding and promotional strategies 3.4 Business strategy development 3.4.1 Vision and Mission statements 3.4.2 SMART Goal 3.5 Building team for startup 3.5.1 Strategies for recruiting founders and early employees 3.6 Financial analysis 3.6.1 Revenue streams and cost structures 3.6.2 Key financial statements for future financial projections 3.6.3 Raising capital for startup: seed capital, angel investment, venture capital
To enhance the knowledge/skills of the students on developing business canvas and business plan and presenting idea deck	**Unit IV: Business Modeling, Planning and Pitching (6 Hrs.)** 4.1 Business canvas 4.1.1 Types of business canvases: business model, value proposition and lean 4.2 Business plan 4.2.1 Importance of the business plan 4.2.2 Contents of a successful business plan 4.3 Business pitch 4.3.1 Contents of successful idea pitch for different stakeholder 4.3.2 Pitch delivery skills
To make the students familiar with professional ethics, roles and responsibilities	**Unit V: Ethics and Responsibilities for Professional Practice (4 Hrs.)** 5.1 Professional ethics and responsibilities 5.1.1 Professional roles and responsibility 5.1.2 Conflict of interest 5.1.3 Whistleblowing 5.1.4 Corporate social responsibility 5.2 Code of conduct-Nepal Engineering Council, IEEE and ACM

To make the students knowledgeable about legal issues on intellectual properties, ICT and business registration and taxation	**Unit VI: Legal Issues for Professional Practice (5 Hrs.)** 6.1 Intellectual Properties 6.1.1 Copyright, patent, design, trademark, and trade-secrets 6.1.2 IP Laws 6.2 IT related policies and laws in Nepal 6.2.1 ICT Policy 6.2.2 Electronics Transaction Act 6.2.3 Communication Act 6.3 Legal issues for startup 6.3.1 Business registration in Nepal 6.3.2 Taxation and compliance requirements

4. Methods of Instruction

Main methods of instruction:

1. Lecture, social media videos and guest entrepreneur speaker on course contents
2. Discussion on lecture contents, case studies, assignments and activities
3. Group work on business canvas, business plan and idea deck
4. Individual field visit and case development of local entrepreneurs
5. Presentation of assignments and business deck for seed fund.

5. List of Tutorials

The following tutorial activities of 15 hours should be conducted to cover all the required contents of this course.

S.N.	Tutorials
Unit 1	Tutorial 1: Students review and discuss paper or video case studies. Tutorial 2: Each student interviews a local entrepreneur and develops a short case on journey and challenges on entrepreneurship. Submit case.
Unit 2	Tutorial 3: Each student identifies a local problem and develops a solution (business idea) and lists the process for startup. Discuss selected ideas.
Unit 3	Tutorial 4: Students explore in-depth issues (based on content of unit 3) for an innovative business idea. Discuss issues of selected ideas. Tutorial 5: Learn company registration process and filling mock registration form Tutorial 6: Discussion on mock loan application form
Unit 4	Tutorial 7: Develop and submit one page business, value or lean canvas. Tutorial 8: Preparation of business idea pitch. Pitch of selected ideas. Tutorial 9: Preparation and submission of detailed business plan. Tutorial 10: Chart paper or banner presentation and gallery walk session based on business canvas, idea pitch and plan.
Unit 5	Tutorial 11: Case studies review and discussion.
Unit 6	Tutorial 12: Group discussions on legal policies and acts.

6. Evaluation system and Students' Responsibilities

Evaluation System

In addition to the formal exam(s) conducted by the Office of the Controller of Examination of Pokhara University, the internal evaluation of a student may consist of class attendance, class

participation, assignments, presentations, written exams, etc. The tabular presentation of the evaluation system is as follows.

External Evaluation	**Marks**	**Internal Evaluation**	**Marks**
Semester-End Examination	50	Class attendance and participation	5
		Case development	5
		Assignments, business canvas, and presentations	5+5+5
		Business Plan	15
		Internal Term Exam	10
Total External	50	Total Internal	50
Full Marks 50+50= 100			

Students' Responsibilities:

Each student must secure at least 45% marks in the internal evaluation with 80% attendance in the class to appear in the Semester End Examination. Failing to obtain such a score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear in the End-Term examinations. Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during the period. If a student fails to attend a formal exam, quiz, test, etc. there won't be any provision for a re-exam.

7. Prescribed Books and References

Reference books:

1. Barringer, B.R., and Ireland, R. D., *Entrepreneurship: Successfully Launching New Ventures*. Pearson.
2. Hisrich, R.D., Manimala, Mathew J., Peters, M. P. and Shepherd, D. A. *Entrepreneurship*. McGraw Hill.
3. Holt, D.H. *Entrepreneurship New Venture Creation*. Prentice-Hall.
4. Timmons, Jeffry A., Spinelli Jr., Stephen. *New venture creation : entrepreneurship for the 21st century*. McGraw-Hill.

Recommended resources:

1. Entrepreneurship and Professional Practice Workbook
2. National and International Policies and Laws
3. Online book and information on technopreneurship, entrepreneurship and professional practice.

Pokhara University
Faculty of Science and Technology

Course No.: CMP 328 (3 Credit)	Full marks: 100
Course title: IT Architecture (**3-1-2**)	Pass marks: 45
Nature of the course: Theory & Practical	Total Lectures: 45 hrs.
Level: Bachelor	Program: Computer/Software/IT

1. Course Description

This course provides an in-depth understanding of IT Architecture principles, methodologies, and best practices. It covers the design, implementation, and management of robust IT infrastructures that align with business goals. Students will learn about various architectural frameworks, enterprise architecture, service-oriented architecture (SOA), and cloud architecture.

2. General Objectives

- Understand the fundamental concepts of IT architecture.
- Analyze and design IT architectures that align with business strategies.
- Apply architectural frameworks and methodologies in real-world scenarios.
- Evaluate and select appropriate technologies and tools for IT architecture.
- Design and implement scalable, secure, and efficient IT infrastructures.

3. Methods of Instruction

Lecture, Discussion, Readings, Practical works, Case Studies and Project works.

4. Contents in Detail.

Specific Objectives	Contents
Understand the IT Architecture with key concepts, evolution and IT Architecture Frameworks.	**Unit 1: Introduction to IT Architecture (9 Hours)** 1.1 Overview of IT System Security: Goals and Importance 1.2 Definition and Importance of IT Architecture 1.3 Evolution of IT Architecture 1.4 Key Concepts and Terminologies 1.5 TOGAF (The Open Group Architecture Framework) 1.6 Zachman Framework 1.7 FEAF (Federal Enterprise Architecture Framework) 1.8 Comparing Frameworks 1.9 IT Architect Roles
Understand the key concept of Enterprise Architecture and Service Oriented Architecture	**Unit 2: Enterprise Architecture and Service Oriented Architecture (8 Hours)** 2.1 Understanding Enterprise Architecture 2.2 Roles and Functions of Enterprise Architecture 2.3 Components of EA: Business, Data, Application, and Technology Architectures 2.4 EA Development Processes and Tools 2.5 Principles of SOA 2.6 Designing and Implementing SOA 2.7 Web Services and Microservices 2.8 Case Studies and Real-world Applications
Understand the idea of business architecture along with security in software development and business architecture modeling	**Unit 3: Business Architecture (7 Hours)** 3.1 Understanding the Business 3.2 Security and Software Development 3.3 Agility in Software Architecture 3.4 Software Development Methods and Practices 3.5 Business Architecture Modeling 3.6 Designing the Business Logic Architecture 3.7 Designing the Business Operation Architecture 3.8 Interoperable Data Systems 3.9 API Management and Integration
Understand the concept of IT Infrastructure Architecture	**Unit 4: IT Infrastructure Architecture (7 Hours)** 4.1 Network Architecture 4.2 Data Center Architecture 4.3 Virtualization and Containerization 4.4 Storage Solutions and Strategies

Understand the concept of security architecture in IT Architecture	**Unit 5: Security Architecture (7 Hours)** 5.1 Principles of Security Architecture 5.2 Designing Secure IT Architectures 5.3 Identity and Access Management (IAM) 5.4 Risk Management and Compliance
Understand the concept of cloud architecture	**Unit 6: Cloud Architecture (7 Hours)** 6.1 Cloud Computing Models: IaaS, PaaS, SaaS 6.2 Designing Cloud Architectures 6.3 Cloud Security and Compliance 6.4 Migration Strategies and Best Practices 6.5 Edge Computing Future Trends and Innovations

5. **Practical Works**

The practical works of 30 hours per group of maximum 24 students should cover the following lab works:

SN	Practical Works
1.	Introduction to Enterprise Architecture Tools (Hands on with TOGAF Tools E.g. Archi : The Open Source modelling toolkit for creating ArchiMate models and sketches used by Enterprise Architects everywhere.)
2.	Designing a simple IT Architecture (Creating a high-level blueprint that outlines the key components, relationships, and technologies required to support an organization's business processes, ensuring scalability, security, and efficiency. This typically includes defining the structure of the network, data flow, application integration, and infrastructure layout to meet current and future business needs.)
3.	Implementing Service Oriented Architecture (Design and deploy services as independent, reusable components that communicate over a network using standardized protocols (e.g., REST, SOAP), ensuring each service performs specific business functions and can be integrated with other services seamlessly.)
4.	Cloud Architecture LAB in AWS (Deploying a Cloud Architecture LAB in AWS involves several steps, including setting up the AWS environment, creating and configuring virtual private clouds (VPCs), launching instances, and deploying necessary services such as ELB, S3 Buckets, Auto Scaling)
5.	Security Architecture Implementation with IAM

6. List of Tutorials:

The various tutorial activities that suit this course should cover all the content of this course to give students a space to engage more actively with the course content in the presence of the instructor. Students should submit tutorials as assignments or class-works to the instructor for evaluation. The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover the content of this course:

A. **Discussion-based Tutorials:** (4 hrs.)

1. Discuss the importance of Enterprise Architecture and Service Oriented Architecture.

B. **Problem solving-based Tutorials:** (6 hrs.)

1. Design a simple IT Architecture of an Enterprise.
2. Develop a Service Oriented Architecture (SOA) of an Enterprise

C. **Review and Question/Answer-based Tutorials:** (5 hrs)

1. Case study on Infrastructure and Cloud Architecture of an Enterprise.
2. Students ask questions within the course content and assignments and review key course content in preparation for tests or exams.

7. **Evaluation system and Students' Responsibilities**

Internal Evaluation

The internal evaluation of a student may consist of assignments, attendance, internal assessment, lab reports and project work etc. The internal evaluation scheme for this course is as follows:

<table>
<tr><th>Internal Evaluation</th><th>Weight</th><th>Marks</th><th>External Evaluation</th><th>Marks</th></tr>
<tr><td>Theory</td><td></td><td>30</td><td rowspan="11">Semester-End examination</td><td rowspan="11">50</td></tr>
<tr><td>Attendance & Class Participation</td><td>10%</td><td></td></tr>
<tr><td>Assignments</td><td>20%</td><td></td></tr>
<tr><td>Presentations/Quizzes</td><td>10%</td><td></td></tr>
<tr><td>Internal Assessment</td><td>60%</td><td></td></tr>
<tr><td>Practical</td><td></td><td>20</td></tr>
<tr><td>Attendance & Class Participation</td><td>10%</td><td></td></tr>
<tr><td>Lab Report/Project Report</td><td>20%</td><td></td></tr>
<tr><td>Practical Exam/Project Work</td><td>40%</td><td></td></tr>
<tr><td>Viva</td><td>30%</td><td></td></tr>
<tr><td>Total Internal</td><td></td><td>50</td></tr>
<tr><td colspan="5">Full Marks: 50 + 50 = 100</td></tr>
</table>

Student Responsibilities:

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Book

1. "*Fundamentals of IT Architecture*" by Daniel Akenine, Eva Kammerfors, Sven-Hakan Olsson, Robert Folkesson

References:

1. "*Enterprise Architecture as Strategy*" by Jeanne W. Ross, Peter Weill, and David Robertson
2. "*SOA Principles of Service Design*" by Thomas Erl
3. "*Architecting the Cloud*" by Michael J. Kavis
4. "*IT Infrastructure Architecture - Infrastructure Building Blocks and Concepts*" by Sjaak Laan
5. Online resources and industry publications

Pokhara University
Faculty of Science and Technology

Course Code: MTH 252	Full Marks: 100
Course title: Numerical methods (2-1-2)	Pass Marks: 45
Nature of the Course: Theory and Practical	Total Lectures: 30 hours
Level: Bachelor	Program: BE

1. Course Description

This course explains how to utilize a computer to solve issues that calculus and algebra might not be able to. It fosters the development of mathematical relationships that can be utilized to model real-world situations and the problem-solving skills necessary to study other engineering courses.

2. General Objectives

The general objectives of this course is to equip students with knowledge and tools required to solve different equations that are applicable in the fields of engineering.

3. Methods of Instructions:

Lecture, Tutorial, Discussion, Readings and Practical works

4. Contents in Detail

Specific Objectives	Contents
Solve non-linear equations by different numerical methods.	Unit 1: **Solution of Non-linear equations (5 hrs)** 1.1. Introduction, Importance of Numerical Methods 1.2. Approximation and Errors in computation 1.3. Bisection Method 1.4. Secant method 1.5. Newton Raphson method 1.6. Fixed point iterative method
Visualize and solve mathematical relationships of practical observations.	Unit 2: **Interpolation and approximation (5hrs)** 2.1. Lagrange interpolation 2.2. Finite differences (forward, backward, and divided difference) 2.3. Newton's Interpolation (forward, backward) 2.4. Least square method of fitting linear and nonlinear curve for discrete data and continuous function 2.5. Cubic Spline Interpolation
Calculate definite integration and differentiation numerically.	Unit 3: **Numerical Differentiation and Integration (4 hours)** 3.1. Numerical Differentiation formulae 3.2. Trapezoidal, Simpson's 1/3, 3/8 rule 3.3. Romberg integration

	3.4. Gaussian integration (2- point and 3- point formula)
Solve the system of linear equations by different techniques.	Unit 4: **Solution of system of linear algebraic equations (6 hours)** 4.1. Gauss elimination method and concept of pivoting 4.2. Ill-conditioned system of linear equations 4. 3. LU Factorization method (Dolittle, Crout's, Cholesky's) 4.4. Iterative methods (Jacobi method, Gauss-Seidel method) 4.5. Eigen value and Eigen vector using Power method
Solve the ordinary differential equations which may exist in the field of engineering.	Unit 5: **Solution of ordinary differential equations (6 hours)** 5.1. Review of ordinary differential equations 5.2. Runge-Kutta methods (first, second and fourth) for first and second order differential equations 5.3. Solution of boundary value problem by shooting method
Solve numerically the partial differential equations which exist in the field of engineering.	Unit 6: **Numerical solution of Partial differential Equation (4 hours)** 6.1. Classification of partial differential equation (elliptic, parabolic and hyperbolic) 6.2. Solution of Laplace equation (standard 5-point formula with iterative methods) 6.3. Solution of Poisson equation (finite difference approximation method) 6.4. Solution of one-dimensional Heat equation by Schmidt method

Note: The figures in the parentheses indicate the approximate periods for the respective units.

5. List of Tutorials

The following tutorial activities of 15 hours per group of maximum 24 students should be conducted to cover all the required contents of this course.

S.N.	List of Tutorials	
1	Determination of a root by all methods and their comparison.	3 hrs
2	Finding of different interpolating polynomials, regression curve and Cubic-spline.	2 hrs
3	Determination of the first and second order derivatives by difference method and its comparison with exact value. Integration by Trapezoid, Simpson's rules, Romberg method, Gaussian method and comparison with exact value.	2 hrs

4	Solution of system of linear equations by Gauss Elimination, matrix factorization, Jacobi, Gauss-seidel method Finding Eigen value and Eigen vector by power method.	4 hrs
5	Solution of first and second order differential equation by RK methods, and Shooting method.	2 hrs
6	Solution of Laplace, and Poisson's equations by five-point formula.	2 hrs

6. List of Practical

SN	List of Practicals
1.	Solution of nonlinear equations.
2.	Interpolation and regression.
3.	Differentiation and Integration.
4.	Linear system of equations and power method.
5.	Ordinary differential equations.

By using MATLAB/C/C++ or any other relevant high level programming languages.

7. Evaluation System and Students' Responsibilities

Evaluation System

The internal evaluation of a student may consist of assignments, attendance, term-exams, lab reports and projects etc. The tabular presentation of the internal evaluation is as follows:

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**30**	**Semester End**	**50**
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		**20**		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		**50**		
Full Marks: 50 + 50 = 100				

Students' Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Books

1. C.F. Gerald and P.O. Wheatley *Applied Numerical Analysis (7th edition),* New york.
2. B. S. Grewal, *Numerical Methods in Engineering and Science* Khanna Publication, (10th edition)
3. S.S. Sastry *Introductory Methods of Numerical Analysis (4th edition),* Prentice-Hall of India, New Delhi, 2008.

References:

1. Richard L. Burden, J. Douglas Faires, "Numerical Analysis 7th edition", Thomson / Brooks/Cole
2. E. Balagurusamy *Numerical methods.* New Delhi; Tata McGraw Hill, 2010.
3. Dr. V. N. Vedamurthy & Dr. N. Ch. S. N. Iyengar *Numerical Methods,* Noida, Vikash Publication House 2009.
4. Rudra Pratap *Getting Started with MATLAB,* Oxford University Press 2010

Sunday, April 20, 2025
8:50 AM

Pokhara University
Faculty of Science and Technology

Course No.: CMP 270	Full marks: 100
Course title: **Research Fundamentals (2-0-2)**	Pass marks: 45
Nature of the course: Theory & Practical	Total Lectures: 45 hrs
Level: Bachelor	Program: BE (Computer, IT and Software)

1. Course Description

This course is designed to develop the skills of students to do a project/research work using the fundamental concepts of research. This course introduces what the research or project work is, explores in brief how it is done, explains what research ethics must be followed during the research/project work and finally guides the students how the research/project documentation is done in the form of the proposals report and final research/project report.

2. General Objectives

- To acquaint the students with basic knowledge of research/project work.
- To develop the skills in students to conduct research/project work.
- To develop the skills in students to work in a team.
- To develop the skills in students to write an impressive proposal report and final research/project report and present their work orally.
- To acquaint the students with the knowledge of research ethics.

3. Methods of Instruction

Lecture, Discussion and Project work.

4. Contents in Detail.

Specific Objectives	Contents

• Understand the basic concepts of research, purpose and outcomes of a research/project work.	**Unit 1: Introduction (4 hrs)** 1.1 What is research? 1.2 Research Aim and Objectives 1.3 Features of Research 1.4 Types of Research 1.5 The 6Ps of Research 1.6 Purpose of Research- reasons for doing research 1.7 Product of Research- outcomes of research 1.8 Research and Project
• Understand and implement the research process model to conduct a research/project work.	**Unit 2: Research Process Model (10hrs)** 2.1 Personal Experiences and Motivation 2.2 Literature Review 1. Purpose and objectives of a literature review 2. Literature resources 3. Conducting a literature review 4. Citation and its types 5. Bibliographic Detail and Referencing Systems 6. Plagiarism 2.3 Research Question 2.5 Conceptual framework 2.5 Strategies 1. Survey 2. Design and Creation 3. Experiment 4. Case Study 5. Action Research and 6. Ethnography 2.6 Data Generation Methods 1. Interview 2. Observations 3. Questionnaire 4. Documents 5. Types of triangulation in a research project 2.7 Data Analysis 1. Quantitative and 2. Qualitative data analysis
• Familiarize with the laws and ethics in research conduction.	**Unit 3: Participants and Research Ethics (4hrs)** 3.1 Participants 3.2 The law and Research 3.3 Rights of People Directly Involved 3.4 Responsibilities of an Ethical Researcher

• Familiarize with the research proposal and its components. • Develop a research/project proposal.	**Unit 4: Proposal Writing** **(4hrs)** 4.1 What is a research proposal? 4.2 Need of a Research Proposal 4.3 Components of a Research Proposal 4.4 A case study on any research paper/project
• Familiarize with the research/project report and its components. • Develop a research/project report.	**Unit 5: Report Writing** **(4hr)** 5.1 What is a research report? 5.2 Need of a Research Report 4.3 Components of a Research Report 4.4 A case study on any research paper/report

5. Practical Work

Laboratory works of 30 hours per group of maximum 24 students should cover all the concepts of research fundamentals studied in the lectures. Students must find a new problem, write a proposal, solve the problem as their project/research work and submit a final project/research report and present their work orally. The marks for the practical work will be based entirely on their project/research work. The entire project/research work shall be divided into two phases and evaluation shall be done accordingly:

Phase I:

- The students are grouped in teams each containing at most 4 students.
- Each team chooses a problem to solve as their project/research work and they work in a team.
- They must define clearly what the problem is, justify why they choose the problem and how they will solve it and submit this as a proposal report (based on Unit 2 and 4).
- Each team presents their proposal orally.

Phase II:

- After the approval of their proposal, they start working on the project.
- Each team follows the research process studied in Unit 2 to do their project/research work.
- Students keep reporting their progress about the project/research work to their instructor.
- Students complete the project/research work, develop the final project/research report (based on Unit 2 and 5) and again present it orally.

6. **Evaluation system**

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		30		
Attendance & Class Participation	10%			
Assignments	20%			

Presentations/Quizzes	10%		Semester-End examination	50
Internal Assessment	60%			
Practical (Project/research Work)		20		
Proposal Report	30%			
Project Presentation	10%			
Final Project Report	40%			
Completeness of Project	20%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

7. Student Responsibilities:

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

1. Oates, B. J., Griffiths, M., & McLean, R. (2022). *Researching information systems and computing*. Sage.
2. Walia, A. M., & Uppal, M. (2020). *Fundamentals of Research.* Notion Press.

8. Annex

A. **Components of Research Proposal**

1. Title Page
2. Abstract
3. Table of Contents, List Figures, List of Tables and Abbreviations
4. Introduction
   a. Rationale/background
   b. Problem and motivation
   c. Aim and objectives of research
   d. Significance of research

   e. Scope of research
   f. Limitation
5. Literature review
6. Research problem and Solution
7. Methodology
   a. Research design
   b. Participants
   c. Data collection methods
   d. Data analysis techniques
   e. Ethical considerations
   f. Validation Techniques
8. Data Analysis and Findings
9. Discussions and Conclusion
10. Contributions and Future Works
11. Reference list/bibliography
12. Annexes

B. Components of Research Report:

1. Title
2. Abstract
3. Keywords
4. Rationale/background and motivation
5. Aim and objectives of research
6. Literature review
7. Research problem
8. Methodology
9. Research plan and budget
10. Contributions- impact and significance
11. Reference list/bibliography
12. Annexes

Other Parts of Research Report:

1. Funding and Acknowledgement
2. Table of Contents, List Figures, List of Tables and Abbreviations
3. Title Page and Copyright Page
4. Declaration and Recommendation
5. Certification Page

***Note**: The components of research proposal and research report may vary based on the research strategy and nature of the research problem.*